

DONGPO LIU

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⚙️ SKILLS

- **Programming Languages:** Experienced in Go/Rust/TypeScript/JavaScript, **multilingual** (can adapt to any language)
- **Development Tools:** familiar with Linux-based programming, have experience with tools like Git, GitHub, Docker, Jenkins, Github Actions and Jira, etc.
- **Languages:** **English** (Can read, write and speak fluently), **Chinese** (Native speaker)

👤 WORK EXPERIENCE

PingCAP Inc. - Database - Database Kernel R&D(Golang/Rust) August 2023 – Present

Transferred from the data platform department to the kernel computing department and started to focus on the development of the optimizer of the distributed database TiDB.

Responsibilities:

- Designed, implemented and maintained the optimizer of the distributed database TiDB.

Accomplishments:

- Designed a statistics update optimization for tables with over 2 billion rows, reducing update time from 718 seconds to 45 seconds and improving performance by 15.8x.
- Designed and implemented a broadcast-based mechanism for cross-node statistics refresh, resolving a long-standing issue that had persisted for five years.
- Optimized the update of statistics for millions of tables, reducing the update duration from 1.5 hours to about 10 seconds.
- Optimized statistics collection by implementing an in-memory priority queue, eliminating repeated queue rebuilding overhead, reducing CPU usage by one core and memory consumption by 99%.
- Designed and implemented the predicate columns feature to optimize statistics collection by only analyzing columns used in predicates, reducing auto analyze time by 50% in OLTP workloads and nearly 99% in partitioned wide table scenarios.
- Designed and implemented the priority queue for speed up the auto analyze process, significantly reducing the starvations of the auto analyze process.
- Added support for the LOCK STATS TABLE feature to allow users to lock statistics updates and generate stable execution plans.

PingCAP Inc. - Database - Database Tools R&D(Golang/Rust) July 2021 – July 2023

Transferred from the community department to the data platform department and started to focus on the development of data synchronization tools TiCDC.

Responsibilities:

- Designed, implemented, and maintained core modules, focusing on Sink and Sorter, and managed daily project maintenance including issue tracking, code reviews, and on-call customer support.

Accomplishments:

- Changed push mode to pull mode to optimize memory management and data synchronization process, greatly improving OOM issues and synchronization lags.
- Modified the Sink module from synchronous to asynchronous mode to optimize code performance and improve throughput by 30% in incremental scenarios.
- Added multi-topic and OAuth authorization support for Kafka Sink, greatly reducing the operation and maintenance costs of multi-topic tasks.
- Added the ability to optimize the Lossy DDL from full table scan to not synchronizing any data, greatly reducing the impact of DDL on the system.
- Refactored all of TiCDC's CLI code to standardize the way commands are added and written, which greatly improved maintainability.

PingCAP Inc. - Database - Full-Stack Engineer(Golang/JS) Aug 2020 – July 2021
Joined the community department to develop TiChi to support community collaboration based on the Kubernetes Community Prow project.

Morningstar, Inc. - Financial Services - Full-Stack Engineer(Java/JavaScript) June 2019 – July 2020

Joined Morningstar, Inc. after graduation, developing and maintaining data presentation components and front-end infrastructure with Java and Vue.js, including user behavior tracking and automated release workflows.

OPEN SOURCE PROJECTS

Cargo(Rust) - Maintainer 301+ commits
The Rust package manager. Cargo downloads your Rust package's dependencies, compiles your packages, makes distributable packages.

- Implemented the 'cargo-info' subcommand to display detailed information about a Rust package.
- Improved various error prompts for Cargo commands and the Rust compiler to make errors to effective and clear.
- Added the -crate-type function and added related documentation for Cargo.
- Added support for automatic inheritance of the [workspace.package] fields for the cargo new/init commands.
- Added new syntax for Cargo build scripts to provide a foundation for future extensions to build scripts.
- Added 'target.'cfg(..).linker' configuration option to allow users to specify the linker via 'cfg(..)' conditions.
- Added OR_PATTERNS_BACK_COMPAT lint to help users migrate to the new orpatterns syntax.
- Made authors field optional in Cargo and rustdoc.

crates.io(Rust/JS) - Previous Maintainer 131+ commits

Rustup(Rust) - Previous Maintainer 110+ commits

Ant Design(TypeScript) - Previous Maintainer 20+ commits

EDUCATION

Software Engineering Bachelor's Degree Sep 2015 – June 2019

- Chongqing University of Technology